

## Supplementary Information

### Environmental impacts of future cotton production in the United States

#### Author Information

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89    **1    System scheme**

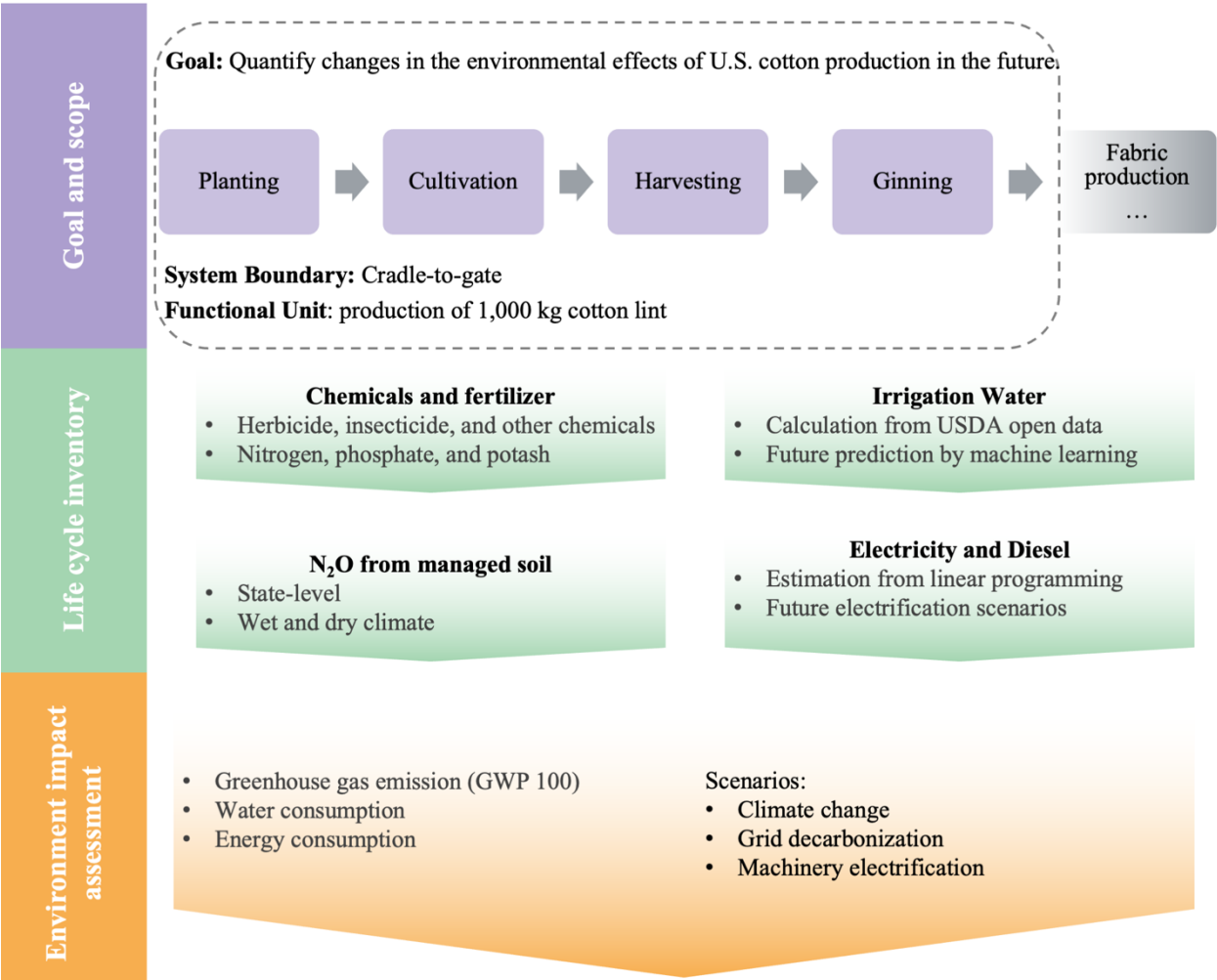


Figure S1. Scheme and system boundary.

## 2 Data collection

The data utilized in this study are categorized into two primary types: cotton agriculture data and climate data. The sources for all datasets are detailed in Supplementary Table 1. All data have been made publicly available and can be accessed via the link: [https://github.com/jenmbdunn/Cotton\\_LCA-main](https://github.com/jenmbdunn/Cotton_LCA-main).

Supplementary Table S1. Data summary

| Data category |                                                      | Data level    | Data range (year)                                 | Data frequency | Data source                                    |
|---------------|------------------------------------------------------|---------------|---------------------------------------------------|----------------|------------------------------------------------|
| Cotton data   | 1. Planted area                                      | State-level   | 2000-2023                                         | Annually       | United States Department of Agriculture (USDA) |
|               | 2. Harvested area                                    | State-level   | 2000-2023                                         | Annually       |                                                |
|               | 3. Yield                                             | State-level   | 2000-2023                                         | Annually       |                                                |
|               | 4. Production                                        | State-level   | 2000-2023                                         | Annually       |                                                |
|               | 5. Irrigated harvested area                          | State-level   | 1997, 2002, 2007, 2012, 2017, and 2022            |                |                                                |
|               | 6. Overall harvested area                            | State-level   | 1997, 2002, 2007, 2012, 2013, 2017, 2018 and 2022 |                |                                                |
|               | 7. Water applied                                     | State-level   | 2003, 2008, 2013, 2018, and 2023                  |                |                                                |
|               | 8. Application of chemicals and fertilizers          | State-level   | 2007, 2015, 2017, and 2019                        |                |                                                |
|               | 9. Costs of fuel, lube, and electricity              | Region-level  | 1997-2023                                         | Annually       |                                                |
|               | 10. Retail price of industry electricity             | State-level   | 2021-2023                                         | Annually       | U.S. Energy Information Administration (EIA)   |
|               | 11. Retail prices for on-highway diesel              | Region-level  | 2020-2023                                         | Annually       |                                                |
| Climate data  | 12. Evaporation                                      | 1.25° x 1.25° | 2000-2014 (History)<br><br>2015-2049 (Projection) | Monthly        | Copernicus Climate Change Service              |
|               | 13. Moisture in the upper portion of the soil column | 1.25° x 1.25° |                                                   | Monthly        |                                                |
|               | 14. Near-surface air temperature                     | 1.25° x 1.25° |                                                   | Monthly        |                                                |
|               | 15. Near-surface specific humidity                   | 1.25° x 1.25° |                                                   | Monthly        |                                                |
|               | 16. Near-surface wind speed                          | 1.25° x 1.25° |                                                   | Monthly        |                                                |
|               | 17. Precipitation                                    | 2.5° x 2.5°   |                                                   | Monthly        |                                                |
|               | 18. Total cloud cover percentage                     | 1.25° x 1.25° |                                                   | Monthly        |                                                |

|  |                  |               |  |         |  |
|--|------------------|---------------|--|---------|--|
|  | 19. Total runoff | 1.25° x 1.25° |  | Monthly |  |
|--|------------------|---------------|--|---------|--|

## 2.1 Cotton agriculture data

The cotton agriculture data were primarily obtained from USDA Quick Stats<sup>1</sup>, a tool provided by the National Agricultural Statistics Service (NASS) of the United States Department of Agriculture (USDA)<sup>2</sup>. The energy-related data, such as electricity prices and diesel fuel prices, were obtained from the U.S. Energy Information Administration (EIA)<sup>3</sup>.

### 2.1.1 Planted area

The dataset focuses on the area planted for cotton, based on an annual survey conducted across 17 U.S. states over a 24-year period (2000–2023). It comprises 408 records, with data values recorded in acres.

### 2.1.2 Harvested area

The dataset focuses on the area harvested for cotton, based on an annual survey conducted across 17 U.S. states over a 24-year period (2000–2023). It comprises 408 records, with data values recorded in acres.

### 2.1.3 Yield

The dataset provides information on annual surveyed cotton yield across the 17 U.S. states, over a 24-year period (2000–2023). It comprises 408 records, with yield values measured in pounds per acre.

### 2.1.4 Production

The dataset provides information on annual surveyed cotton production across the 17 U.S. states, over a 24-year period (2000–2023). It comprises 408 records, with production values measured in 480-pound bales.

### 2.1.5 Census irrigated harvested area

The dataset provides information on irrigated area harvested for cotton, derived from census data collected across various U.S. states. It spans multiple years, like 1997, 2002, 2007, 2012, 2017, and 2022, and focuses on the area harvested under irrigated production practices. It comprises 103 records, representing 17 U.S. traditional cotton cultivation states across separate years, with Illinois included as an exception. Among these, 8 records are marked as "(D)", which means withheld by NASS to avoid disclosing data for individual operations. The value for the irrigation harvested area is recorded in acres.

### 2.1.6 Census overall harvested area

The dataset provides information on overall area harvested for cotton, derived from census data collected across various U.S. states. It spans multiple years and focuses on the area harvested under all utilization practices. It comprises 103 records, representing 17 U.S. traditional cotton



cultivation states across separate years, with Illinois included as an exception. Among these, 3 records are marked as "(D)". The value for overall harvested area is recorded in acres.

#### **2.1.7 Census water applied**

The dataset provides information about water application for irrigated cotton across 17 U.S. states, and Illinois, which we excluded because its share of cotton production is very small. The dataset includes data for the year 2003, 2008, 2013, 2018, and 2023.

#### **2.1.8 Applications (chemical and fertilizer)**

The data obtained from USDA surveys covers the total applications used in the cotton cultivation process, including chemicals such as fungicides, herbicides, insecticides, and others (e.g., limestone). It also includes fertilizers like nitrogen, phosphate, potash, and sulfur. The data spans four separate years—2007, 2015, 2017, and 2019—and includes information from 13 states. The data of applications is measured in pounds.

#### **2.1.9 Energy consumption**

The first dataset used in this section, titled "Cotton Costs and Returns", provides detailed information on the economic and operational aspects of cotton cultivation, derived from USDA surveys. It categorizes costs and returns using the unit "dollars per planted acre", offering insights into various cost components, including operation costs, chemical inputs, and fertilizer usage. For this study, we focus on the "fuel, lube, and electricity" row under operation costs, as these are critical for estimating the energy usage associated with cotton cultivation. The dataset spans 27 years, from 1997 to 2023. It disaggregates U.S. national data into multiple regions including the Heartland, Prairie Gateway, Eastern Uplands, Southern Seaboard, Fruitful Rim, and Mississippi Portal.

The second dataset contains the annual average retail price of electricity for industrial use across each U.S. state from 2001 to 2023, measured in U.S. cents per kilowatt-hour. The data source is the U.S. Energy Information Administration.

The last dataset, also sourced from the EIA, contains the annual retail prices for on-highway diesel spanning consecutive years up to 2024. It includes data for regions such as the East Coast, New England, Central Atlantic, Lower Atlantic, Midwest, Gulf Coast, Rocky Mountain, West Coast, West Coast except California, and California, all measured in U.S. dollars per gallon. See Section 4.6.1 for information on our methodology to convert on-road diesel prices to off-road diesel prices.

### **2.2 Climate data**

The climate data used in this study were obtained from Copernicus Climate Change Service, Climate Data Store<sup>4</sup>. Monthly global climate projections data computed in the framework of the sixth phase of the Coupled Model Intercomparison Project (CMIP6) were utilized.

### 2.2.1 Climate model

The data generated by Community Earth System Model 2 (CESM2) were used in this study<sup>5</sup>. The geographical area covering the contiguous United States has been extracted by isolating the relevant sub-regions within the dataset. Data were collected from January 2000 to December 2014 for the historical dataset. Data from January 2015 to December 2049 are the projection period.

### 2.2.2 Feature descriptions

A total of eight features were included in this study for building machine learning models: evaporation ( $\text{kg m}^{-2} \text{s}^{-1}$ ), moisture in the upper portion of the soil column ( $\text{kg m}^{-2}$ ), near-surface air temperature (K), near-surface specific humidity (dimensionless), near-surface wind speed ( $\text{m s}^{-1}$ ), precipitation ( $\text{kg m}^{-2} \text{s}^{-1}$ ), total cloud cover percentage (dimensionless), and total runoff ( $\text{kg m}^{-2} \text{s}^{-1}$ ). Utilizing the features, the machine learning models are employed to predict future cotton irrigation water demand at the state level. See SI section 4 for additional details.

## 3 Data processing

### 3.1 Cotton agriculture data processing

The datasets were imported from Excel files, with relevant columns (location, year, and value) extracted to streamline the analysis. The value column, containing numerical data with commas as thousand separators, was cleaned by converting it to a numeric format, with invalid values converted to null values. Invalid entries, such as several marked with (D), which indicates data withheld to protect individual operations. To enable year-wise analysis, the datasets were transformed into pivot tables where rows represented locations, columns corresponded to years, and cell values contained the recorded values (e.g., planted area). To ensure continuous data coverage from 2000 to 2023 or 1997 to 2023, missing years were added as new columns, and linear interpolation was applied along the temporal axis to fill the missing data, with extrapolation at the boundaries to maintain consistency. This data preprocessing produced clean, complete datasets suitable for time-series analysis.

### 3.2 Climate data processing

Climate data processing is designed to combine geographic and climate data for selected 17 cotton cultivation states in the contiguous United States. It starts by importing a detailed map of U.S. state boundaries and narrowing the map to include only the relevant 17 states. The geographic data is then standardized to ensure it aligns with other datasets being analyzed. Next, a climate dataset in a NetCDF format that contains information such as precipitation or temperature is loaded. This dataset includes grids of latitude and longitude coordinates, which are adjusted to match the map's geographic reference system. Then a representation of the geographic grid points, associating them with the corresponding climate data is created. For each timestamp (representing a month) in the climate dataset, the climate variable values are matched with the geographic grid points. Figure S2 shows examples of precipitation and runoff for a randomly selected month. These points are then linked to their respective states by identifying which points fall within the boundaries of each state.

The climate data is grouped by state, and the average of these points that fall within the boundary of the state, is calculated and used to represent the monthly climate variable for the state.

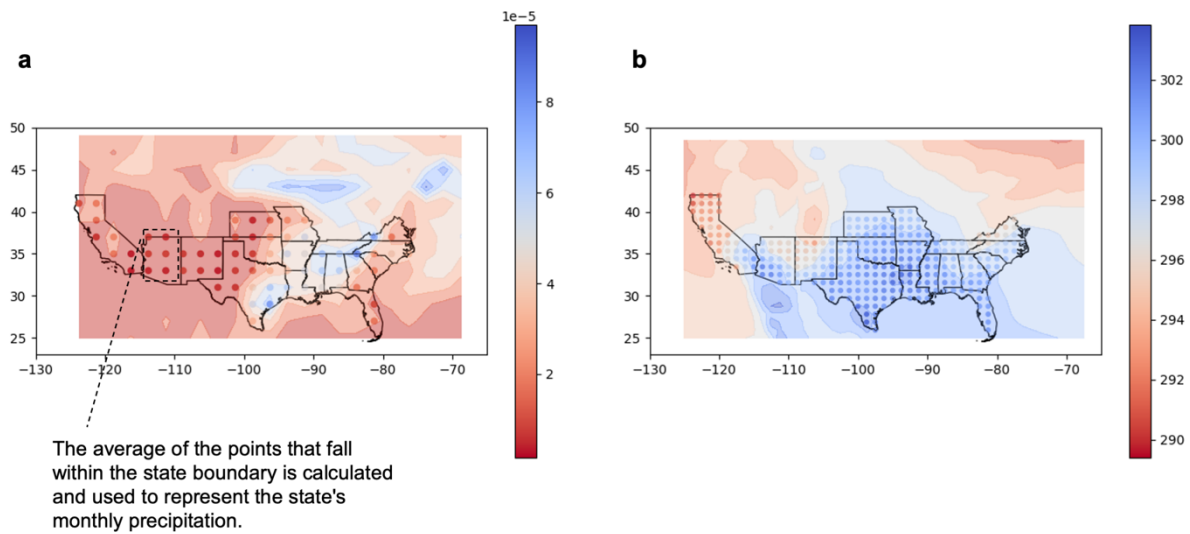


Figure S2. **a** Demonstration of a random month's precipitation in each state. **b** Demonstration of a random month's runoff in each state.

In this study, the average climate data were extracted and used as features in machine learning models. To justify our use of mean climate data in this study, we assessed correlation among minimum, maximum, and mean values of precipitation data. Correlation among them was high, confirming the appropriateness of using the mean value.

Supplementary Table S2. Correlation between minimal, maximum, and mean precipitation data

| Correlation | Mean | Max  | Min  |
|-------------|------|------|------|
| Mean        | 1    | 0.88 | 0.89 |
| Max         |      | 1    | 0.63 |
| Min         |      |      | 1    |

## 4 Data interpretation and method

### 4.1 Planted area, harvested area, and abandonment rate

Figure S2 illustrates the temporal trends in cotton planted area, harvested area, and abandonment rate across major cotton-producing states in the United States from 2000 to 2023. Supplementary Figure S3a depicts the planted area (in acres), while 3b shows the harvested area (in acres) for each state. The data are stacked to highlight the total planted and harvested areas as well as the contributions of individual states. Texas consistently dominates in terms of planted and harvested areas, contributing the largest share throughout the period. Fluctuations in both total planted and harvested areas are notable, with slight declines in later years. The abandonment rate, shown in Figure S3c, representing the proportion of the planted area that was not harvested, is calculated using Equation (S1) below. This abandonment rate is determined by subtracting the ratio of the harvested area to the planted area from 1. Figure S3c uses a bubble chart format. Larger bubbles indicate higher abandonment rate. For instance, states like Texas and Oklahoma show relatively higher abandonment rates, as seen by larger bubbles. Some states experience consistently low abandonment rates, which may reflect stable growing conditions or better management practices. Additionally, abandonment rates in certain years, such as 2011, are significantly high, likely reflecting adverse weather events or other challenges during that time. Furthermore, after 2011, there is an increasing trend in abandonment rates from 2011 to 2023 compared to the previous decade, which may also indicate worsening climate conditions or additional challenges during this period.

$$\text{Abandonment rate} = 1 - \frac{\text{harvested area (acre)}}{\text{planted area (acre)}} \quad (\text{S1})$$

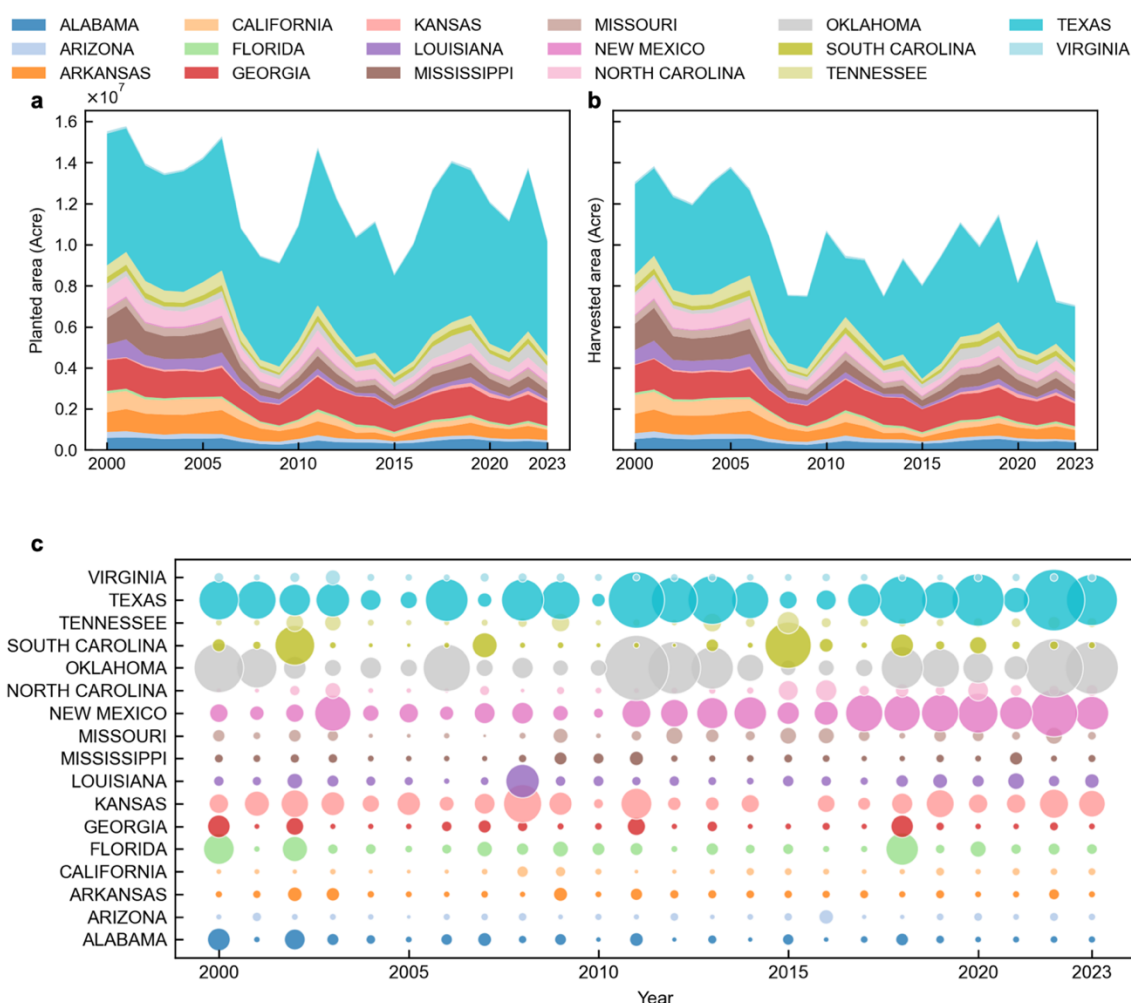


Figure S3. **a** planted area, **b** harvested area, and **c** abandonment rate.

## 4.2 Irrigation rate and water application for irrigated cotton by each state

The resulting irrigation rate values represent the fraction of harvested land that was irrigated, expressed as a ratio ranging from 0 to 1. The calculation utilized census data on irrigated harvested area and overall harvested area, as shown in Equation (S2). Figure S4 illustrates the irrigation rate across 17 states since the year 2000. In the figure, darker colors represent higher irrigation rates. These 17 states are traditionally divided into four production areas based on their geographic location (e.g., Far West, Southeast, Mid-South, and Southwest). When analyzing the irrigation rate data using a hierarchically-clustered heatmap, the states can be grouped into three major categories. These groupings align closely with the four traditional production areas. The Far West region exhibits a high irrigation rate, the Southeast has a low irrigation rate, and the Mid-South and Southwest regions show medium irrigation rates. There are only three exceptions: Arkansas, Tennessee, and Georgia. This suggests that irrigation rate is highly correlated with geographic location.

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$$\text{Irrigation rate} = \frac{\text{irrigated harvested area (acre)}}{\text{overall harvested area (acre)}} \tag{S2}$$

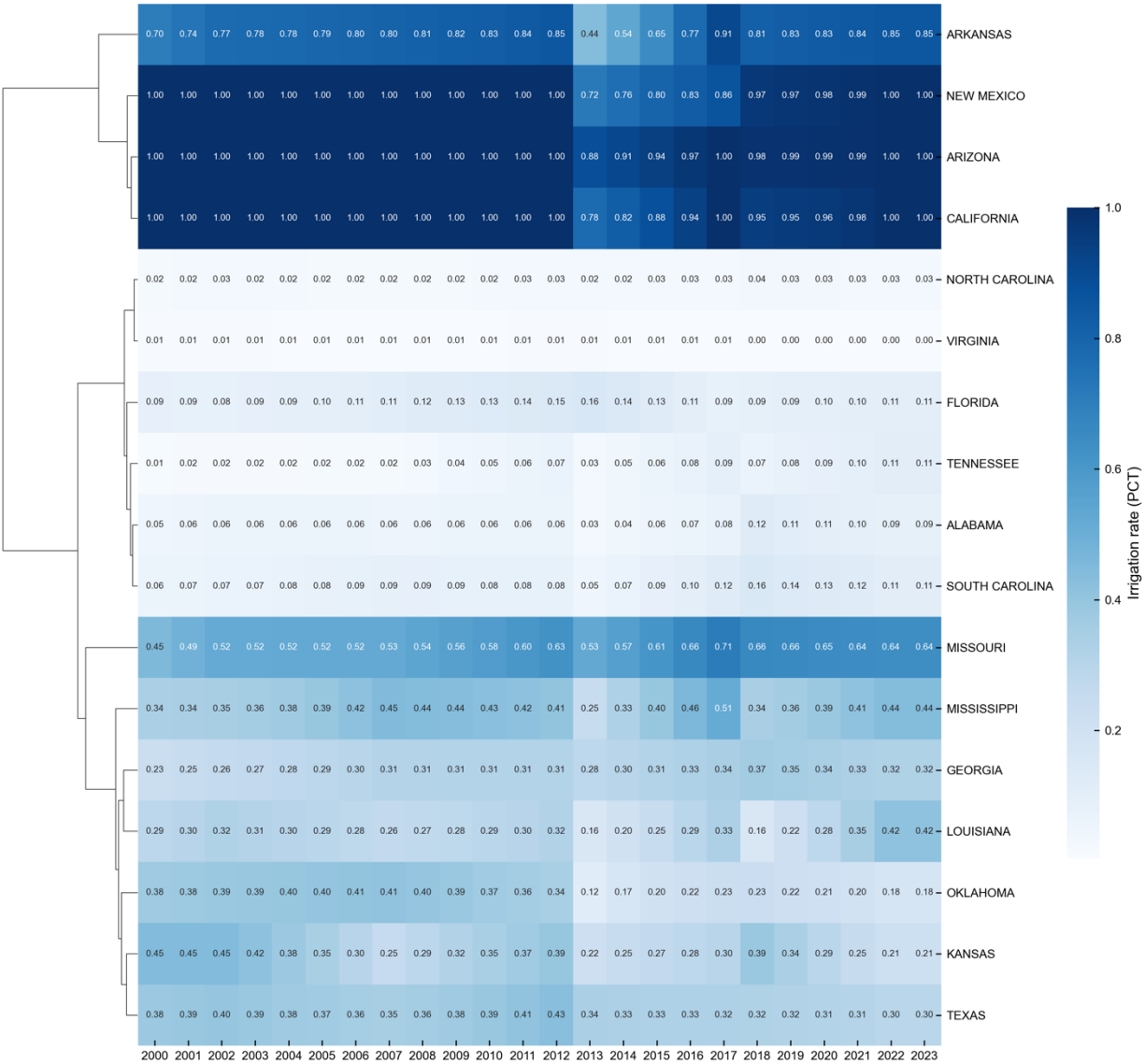
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Figure S5 presents the water applied for irrigated cotton across each state. The illustration is based on census data for water applied, with missing values filled using a linear interpolation method as mentioned in section 3.1. Arizona and California are the two states using significantly more water for irrigated cotton compared to other states.



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Figure S4. Irrigation rate in 17 states 2000-2023

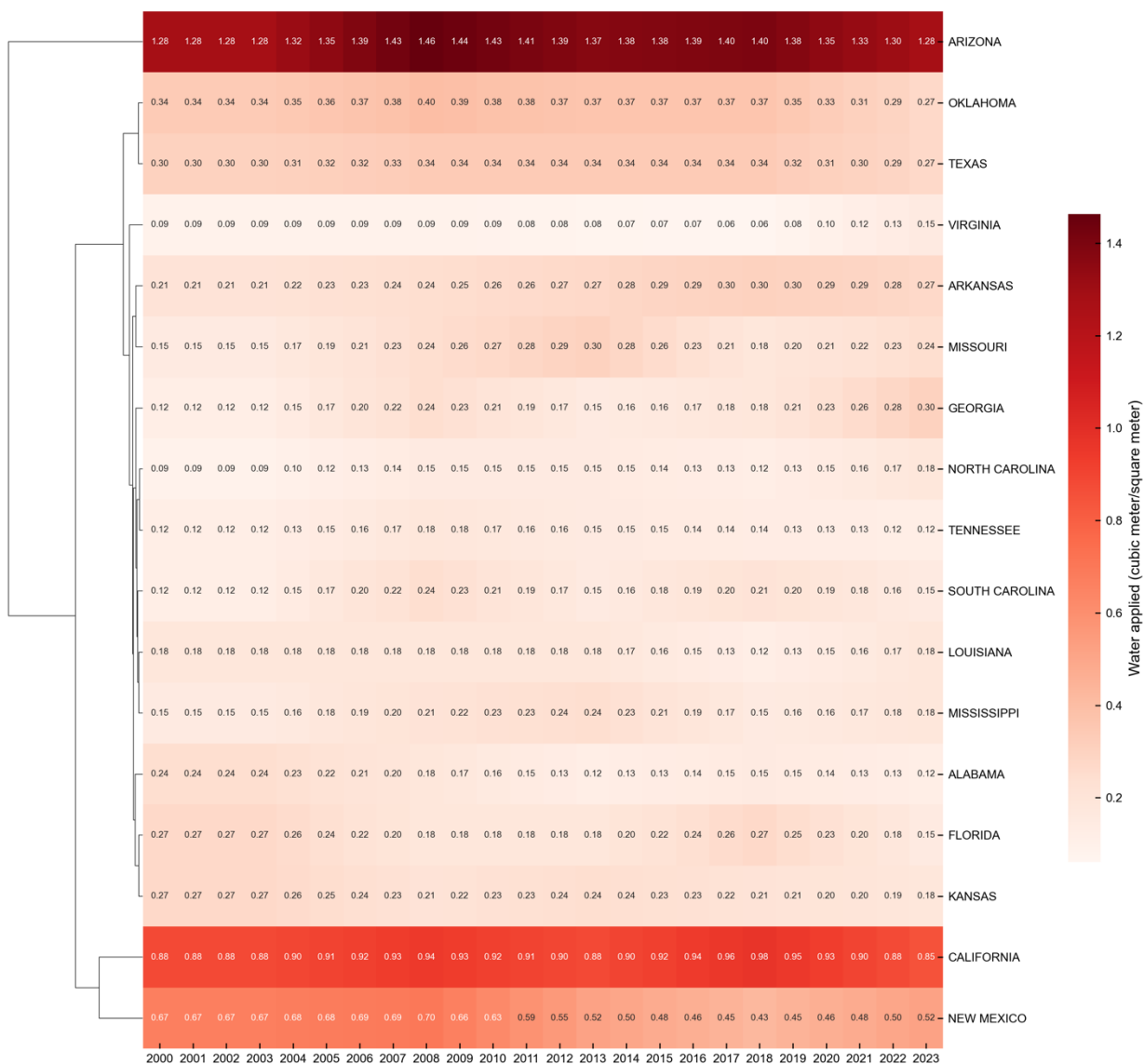


Figure S5. Water applied ( $\text{m}^3/\text{m}^2$ ) for irrigated cotton in 17 states 2000-2023.

### 4.3 Production by each state

Figure S6 shows the annual percentage contribution of each state to total cotton production in the United States from 2000 to 2023, represented as a stacked bar chart. A dashed line indicates total production in kilograms, shown on a secondary y-axis (on the right). The original production data is reported in bales, a standard unit used in cotton production. To enable calculations in metric units, the data were converted from bales to kilograms using the following equivalence: 1 bale = 480 pounds = 217.724 kilograms. States such as Texas and Arkansas consistently contribute the largest shares to total cotton production, whereas states like Arizona and Alabama have smaller contributions. The total production, as shown by the dashed line, exhibits fluctuations over time, with peaks around 2005.

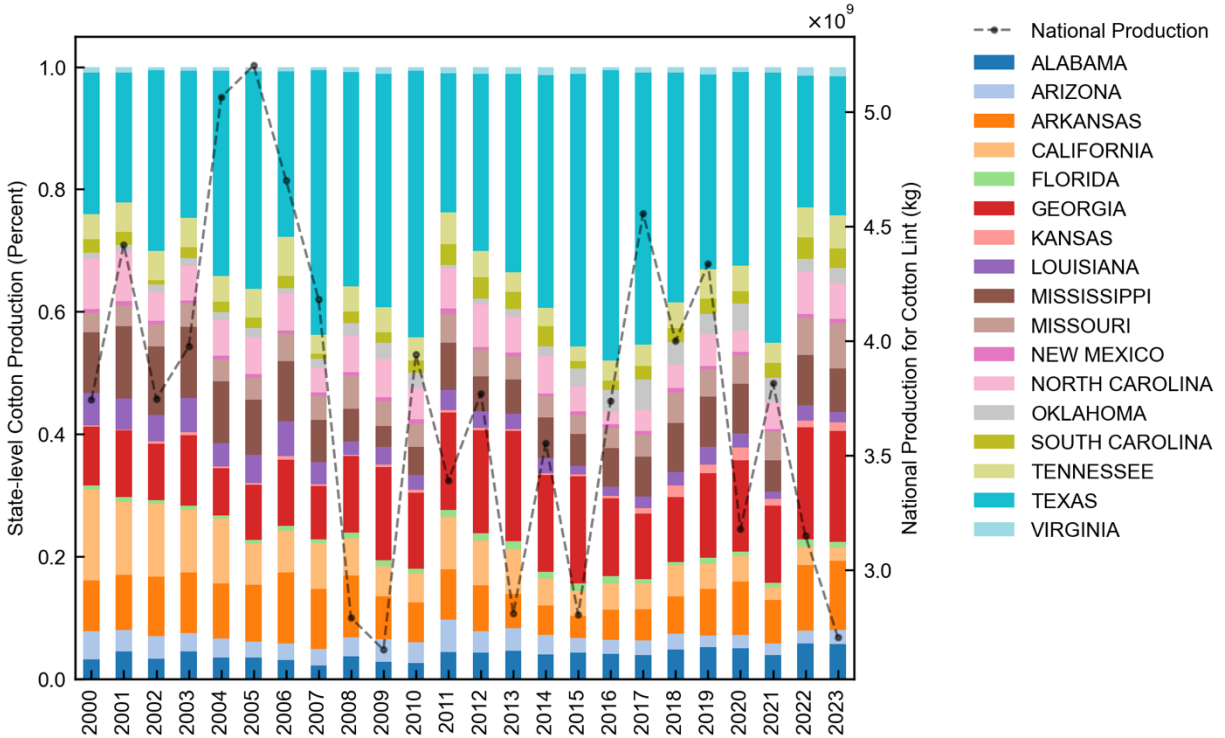


Figure S6. Cotton production in 17 states 2000-2023

#### 4.4 Total irrigation water consumption by each state

The total irrigation water consumption for cotton is estimated by multiplying the planted area, the irrigation rate, and the water applied per acre, as shown in Equation (S3). The original units for planted area were acres, and water applied was reported in acre-feet per acre. To express total water consumption in metric units, the values were converted from acre-feet to cubic meters using the conversion factor: 1 acre-foot = 1233.48 cubic meters.

$$\text{Irrigation water consumption} = \text{harvested area} \times \text{irrigation rate} \times \text{water applied} \quad (\text{S3})$$

As defined in Equation (S2), the irrigation rate used in this analysis was calculated as the ratio of the irrigated harvested area to the overall harvested area.

Figure S6 illustrates the annual percentage contribution of each state to total water consumption for cotton production in the United States from 2000 to 2023. The stacked bar chart represents the proportion of water consumed by each of the 17 states. The primary y-axis (on the left) shows the percentage contribution of each state to total water consumption. The dashed line indicates the total water consumption in cubic meters on a secondary y-axis. The secondary y-axis (on the right) represents the total water consumption in cubic meters. Texas contributes a consistently high percentage to total water consumption throughout the years. The dashed line shows an overall gradual decline in total water consumption over time, with peaks observed in the early 2000s and



around 2011. This figure highlights the temporal and spatial variability in water consumption across states.

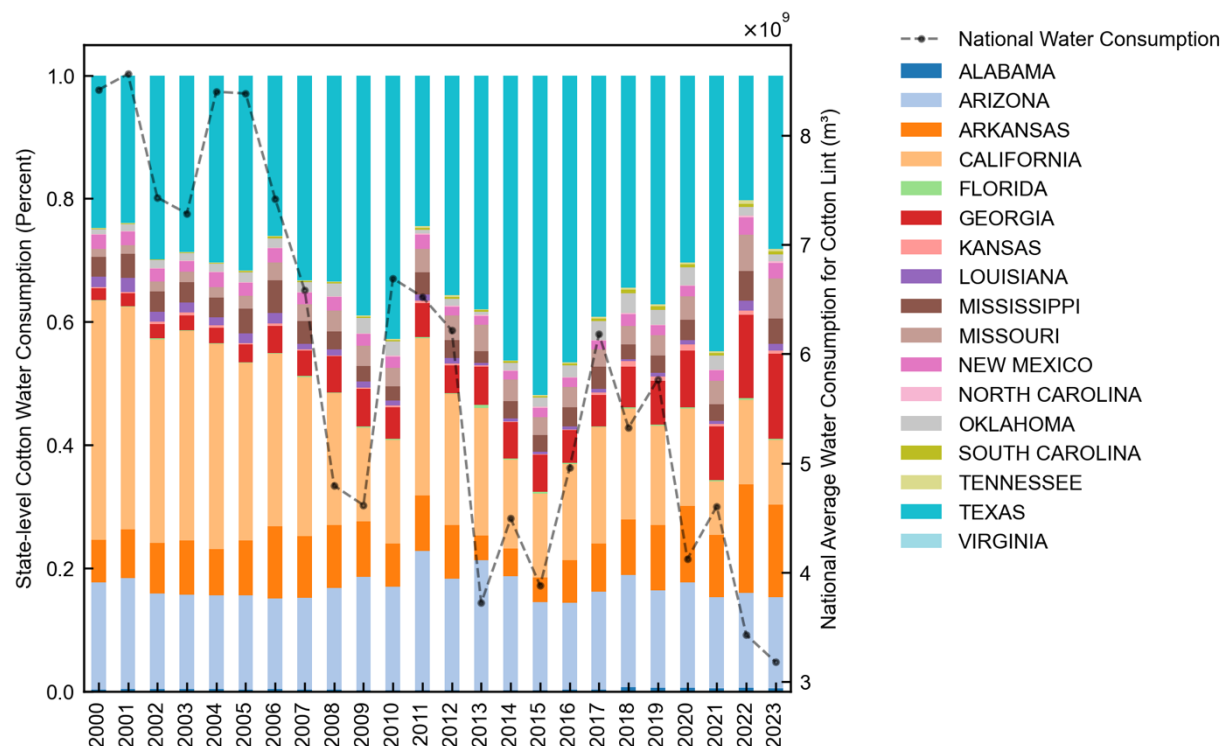


Figure S7. Cotton water consumption in 17 states 2000-2023.

#### 4.5 Unit water consumption in each state

The raw data for water consumption is reported as total water consumed for cotton agriculture in one state in a single year. Ideally, we could use mass allocation to spread the burden of irrigation across the two products from a cotton field: cotton seed and cotton lint. To accomplish that allocation, we would divide the volume of irrigation water by the total amount of mass produced of cotton seed and cotton lint in that state and year. However, data for the mass of cotton seed is not available. Accordingly, we allocate the amount of irrigation water used for cotton lint by multiplying the total irrigation water by the share of gross market value of cotton agricultural products attributable to cotton lint, which is 0.824 based on data from 1997 to 2023. This allocated volume of water is then divided by the total mass of cotton lint produced in the state in that year. Equation (S4) calculates the irrigation water intensity for cotton lint based on this methodology.

$$\text{Unit water consumption} \left( \frac{\text{m}^3}{1000 \text{ kg}} \right) = \frac{\text{allocation parameter} \times \text{total irrigation water consumption} (\text{m}^3)}{\text{production (kg)}} \times 1000 \quad (\text{S4})$$

## 4.6 National average unit water consumption

The national average unit water consumption shown in Equation (S5) is a weighted average of unit water consumption at the national level, using data from 17 states. It is expressed in cubic meters (m<sup>3</sup>) water consumption per 1000 kilograms (kg) of cotton lint.

$$\text{National unit water consumption} \left( \frac{\text{m}^3}{1000 \text{ kg}} \right) = \frac{\text{allocation parameter} \times \sum_{i=0}^{17} \text{total irrigation water consumption (m}^3\text{)}}{\sum_{i=0}^{17} \text{production (kg)}} \times 1000 \quad (\text{S5})$$

The summation  $\sum_{i=0}^{17}$  total irrigation water consumption (m<sup>3</sup>) aggregates the water usage across 17 states. The summation  $\sum_{i=0}^{17}$  production (kg) represents the total production across all 17 states. National level energy consumption estimation

### 4.6.1 Electricity and diesel consumption

On-highway diesel price dataset collected in this study is the average retail price for diesel used in motor vehicles. The key differences between farm used diesel (off-road diesel) and on-highway diesel (No. 2 diesel) lie in their taxation. Off-road diesel is exempt from federal and most state excise taxes, making it significantly cheaper. In contrast, highway diesel is subject to federal excise tax (currently \$0.244 per gallon) and state taxes. As of July 1, 2024, state taxes and fees on diesel fuel averaged \$0.3476 per gallon of diesel fuel<sup>6</sup>. In this study, the off-road diesel price was estimated by using the Equation (S6).

$$\text{Off road diesel price} = \text{on highway diesel price} - \text{state tax} - \text{federal tax} \quad (\text{S6})$$

The federal tax on diesel fuel has remained stable and has not been adjusted for inflation since 1993. Meanwhile, the average state tax on diesel fuel increased from \$0.1929 in 2000 to \$0.3476 in 2024<sup>7</sup>. Missing average state tax on diesel fuel for the years 2021, 2022, and 2023 were estimated using linear interpolation, based on recorded data from 2020 (\$0.2960) and 2024 (\$0.3476). The annual average retail price of electricity for industrial use across each U.S. state from 2001 to 2023, measured in U.S. cents per kilowatt-hour, was also used in this section to estimate energy consumption.

### 4.6.2 Electrification scenario

The energy content of diesel fuel is approximately 128,450 btu per gallon based on GREET, equivalent to 9.94 kWh per liter. Considering the thermal efficiencies of diesel and electric engines, which are typically 40% and 90%<sup>8,9</sup>, respectively, the useful energy output from 1 liter of diesel is calculated as follows: Useful Energy (Diesel)=9.94 kWh/L×0.40=3.976 kWh/L. To achieve the same useful energy output with an electric engine, the required electricity is determined by dividing the useful energy by the efficiency of the electric engine: Required Electricity=3.976/0.90≈4.42 kWh. Therefore, a minimum of 4.4 kWh of electricity is required to replace the energy provided by 1 liter of diesel. This estimation is conservative compared to a previous study based on the California Air Resources Board's Energy Economy Ratio (EER) values for electrified heavy-duty off-road vehicles<sup>10,11</sup>.

These collected data do not separately report diesel and electricity consumption, so we solved a series of linear equations that make up the lines in Figure 4a to identify a reasonable conversion factor between diesel and electricity consumption. After completing the coefficients of linear equations in Equation (2) in main text, pairwise intersections were calculated to identify potential solutions for diesel and electricity usage combinations. Intersections outside the operational limits of  $0 \leq x \leq 500$  (diesel usage) and  $0 \leq y \leq 5000$  (electricity usage) were filtered out. These remaining intersections, representing feasible solutions, were plotted. The purple intersections shown in Figure S8 are the solutions to pairs of the series of linear equations. The green line is the best linear fit of these solutions. This function can be used as the conversion function between diesel and electricity quantities in practice, subject to the total cost of energy, which indicates that replacing 1 litre of diesel with 11 kWh of electricity is equivalent. The dashed yellow line accounts for the relationship between the efficiency of a diesel engine versus electric farm equipment. It represents a proposed electricity only scenario, illustrating the trade-off between diesel and electricity usage, subject to the output work. The point of intersection of these two lines is the set of  $x$  and  $y$  values we use to convert diesel consumption to electricity consumption, thereby enabling analysis of a scenario with purely electric equipment. The representative intersection (130.6 L diesel/1,000 kg cotton lint and 784.4 kWh electricity/1,000 kg cotton lint), marked in red, has the closest distance to the others and was thus chosen as the estimate for the energy usage of cotton cultivation in this study. The representative intersection is also within the reference ranges for diesel and electricity usage<sup>12</sup>, which indicates an appropriate estimation. The yellow dashed line lies within the upper part of the red line, implying that replacing 1 litre of diesel with 4.4 kWh of electricity can not only meets the output work requirements but also adheres to the cost constraints. The electricity usage upper bound and electricity usage lower bound shown as Figure S8 were defined as electricity use in relation to zero diesel consumption for producing 1000 kg of cotton lint. The upper bound (worst case) derived from the fitted relationship between electricity and diesel usage (e.g.,  $y = -11.0x + 2211.0$ ). The lower bound (best case) come from the theoretical calculation of electricity requirements (e.g., using the  $y = -4.4x + 1359$  pro-electricity scenario equation). The electricity usage upper bound and electricity usage lower bound were then used for emission estimation.

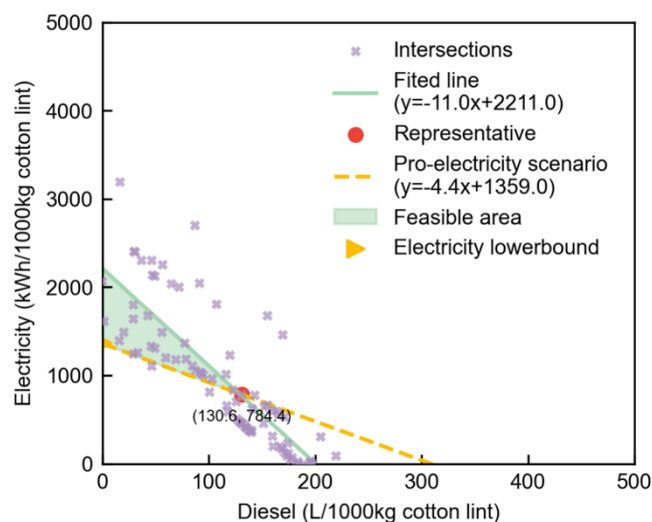


Figure S8. Estimation of current energy usage and future electrification scenario

To access the impact of evolving of U.S. electricity mix, the following time series change in GREET model were implemented as shown in Figure S9.

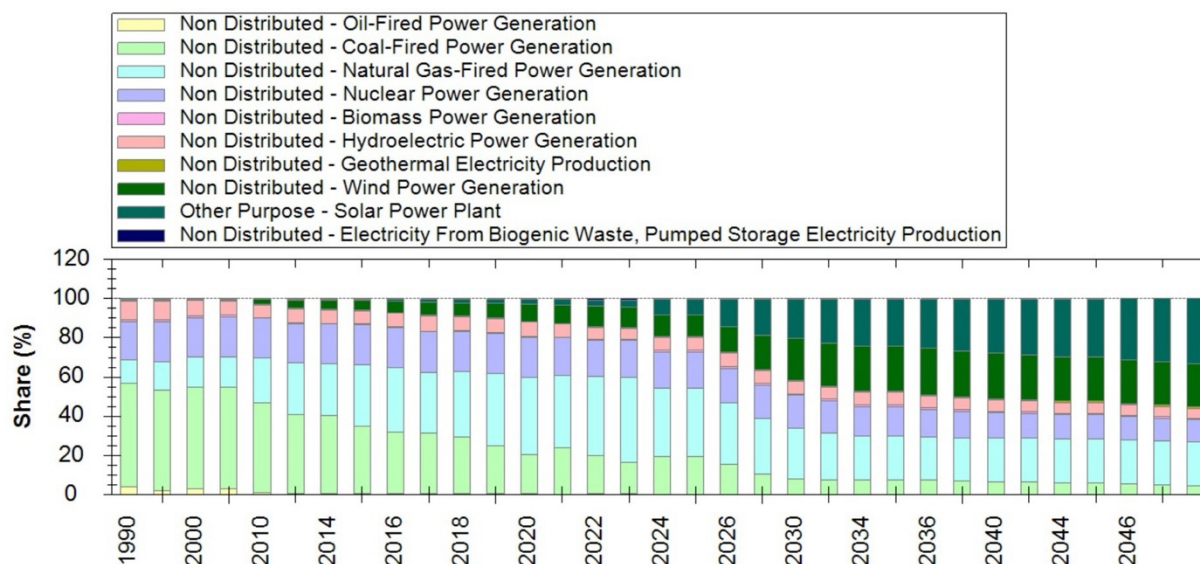


Figure S9. Evolution of U.S. electricity mix <sup>13</sup>.

#### 4.7 Chemical and fertilizer consumption estimation

This section quantifies chemical and fertilizer usage in cotton production in the United States using data from 2015 and 2019, as these years are recent and have minimal missing data. The methodology involves allocating chemical and fertilizer usage to cotton lint and calculating both state-level and national averages based on cotton production data. Production data for cotton was extracted for 2015 and 2019. Since the dataset for chemical usage covers only 13 states, production

data for 4 states out of the 17 were removed to maintain consistency. For each chemical and fertilizer category, including herbicides, insecticides, other chemicals (mainly calcium carbonate), nitrogen, phosphate, potash, and sulfur, the usage was normalized by the production data at state level as shown in Equation (S7). Allocation parameter for cotton lint equals 0.824.

$$\text{Normalized chemical and fertilizer usage} = \frac{\text{chemical and fertilizer usage}}{\text{production}} \times \text{allocation parameter} \quad (\text{S7})$$

Thirteen cotton-producing states were analyzed, and their chemical and fertilizer usage contributions in 2019 were displayed in main text. The states included Alabama (AL), Arizona (AZ), Arkansas (AR), California (CA), Georgia (GA), Louisiana (LA), Mississippi (MS), Missouri (MO), North Carolina (NC), Oklahoma (OK), South Carolina (SC), Tennessee (TN), and Texas (TX).

To determine the national average chemical and fertilizer usage for each category, a weighted average approach was adopted. Chemical and fertilizer usage contributions of each state in 2019 were used, as 2019 is the most recent year with the fewest missing values in the obtained original data. The national average usage for each chemical and fertilizer was calculated as Equation (S8). The calculated averages for herbicides, insecticides, other chemicals, nitrogen, phosphate, potash, and sulfur were presented in main text.

$$\text{National average usage} = \frac{\sum_{i=1}^{13} \text{Normalized chemical and fertilizer usage} \times \text{state production}}{\sum_{i=1}^{13} \text{state production}} \quad (\text{S8})$$

404 **4.8 State-level pro-electricity scenario**

405 Supplementary Table S3. Summary of diesel and electricity consumption for each state.

| State          | Diesel consumption<br>(L/1,000kg cotton lint) | Electricity consumption<br>(kWh/1,000kg cotton lint) | Observed diesel<br>convert to<br>electricity rate | Theoretical<br>Substitution<br>Rate<br>(kWh/L<br>Diesel) |
|----------------|-----------------------------------------------|------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| Alabama        | 33.7                                          | 414.1                                                | 7.8                                               | 4.4                                                      |
| Arizona        | 95.6                                          | 1101.1                                               | 11.3                                              |                                                          |
| Arkansas       | 71                                            | 490.4                                                | 12.6                                              |                                                          |
| California     | 94.9                                          | 574.3                                                | 4.6                                               |                                                          |
| Florida        | 110.2                                         | 463.1                                                | 5.7                                               |                                                          |
| Georgia        | 56.4                                          | 741.2                                                | 9.1                                               |                                                          |
| Kansa          | 101.3                                         | 1172.8                                               | 7.3                                               |                                                          |
| Louisiana      | 82.4                                          | 523.5                                                | 12.6                                              |                                                          |
| Mississippi    | 77.5                                          | 465.7                                                | 12.5                                              |                                                          |
| Missouri       | 93.8                                          | 746.2                                                | 8.3                                               |                                                          |
| North Carolina | 75.5                                          | 994.9                                                | 10.0                                              |                                                          |
| New Mexico     | 199.9                                         | 2082.1                                               | 7.4                                               |                                                          |
| Oklahoma       | 114.3                                         | 1214.6                                               | 5.0                                               |                                                          |
| South Carolina | 65.2                                          | 729.8                                                | 9.6                                               |                                                          |
| Tennessee      | 26.1                                          | 423.7                                                | 8.5                                               |                                                          |
| Texas          | 189.2                                         | 1191.7                                               | 5.4                                               |                                                          |
| Virginia       | 48.7                                          | 677.9                                                | 10.3                                              |                                                          |

## 5 Machine learning models

This study employs machine learning algorithms to predict future irrigation water usage for cotton from feature sets derived from climate data.

### 5.1 Model selection

The study employs a combination of machine learning models to enhance predictive accuracy and robustness. The models include Multi-Layer Perceptron (MLP), Random Forest Regressor (RF), K-Nearest Neighbors Regressor (KNN), and a Stacking Regressor for ensemble learning.

The MLP is selected due to its capability to capture complex nonlinear relationships within the dataset, which is essential for modeling intricate dependencies in the input variables<sup>14,15</sup>. The Random Forest Regressor (RF) is incorporated as a tree-based ensemble model that aggregates multiple decision trees through bootstrap sampling, thereby reducing variance and enhancing generalization<sup>16</sup>. The RF model is optimized using hyperparameter tuning, including variations in the number of estimators and the minimum sample requirements for node splits and leaf nodes. The primary motivation for employing Random Forest lies in its ability to handle structured data efficiently while being resistant to overfitting. The K-Nearest Neighbors (KNN) Regressor is employed as a non-parametric, instance-based learning algorithm that determines predictions based on the proximity of data points in the feature space<sup>17,18</sup>. The KNN model is optimized by varying the number of neighbors, weighting schemes, and distance metrics. The inclusion of KNN is justified by its ability to capture local patterns in the data distribution, making it a valuable benchmark for comparison against other regression models.

To leverage the strengths of individual models and improve overall prediction accuracy, a Stacking Regressor is implemented as an ensemble learning approach<sup>19</sup>. The stacking model integrates multiple MLP-based pipelines as base learners and employs a constrained linear regression model as the final meta-estimator. By learning an optimal weighting of the predictions from individual MLP models, the Stacking Regressor enhances generalization while mitigating the risk of overfitting. The choice of a linear regression model as the meta-learner ensures an interpretable and balanced combination of base model predictions.

The integration of these diverse machine learning approaches, encompassing neural network, ensemble learning, and non-parametric methods, ensures a comprehensive and robust predictive framework. The comparative evaluation of these models facilitates the identification of the most suitable approach for the given dataset while enhancing the reliability of the results.

### 5.2 Model evaluation criteria

The model performance is evaluated using three statistical error metrics: the coefficient of determination ( $R^2$ ), mean absolute percentage error (MAPE), and root mean squared error (RMSE). The mathematical definitions of these metrics are provided below.

The coefficient of determination is computed as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (S9)$$

where  $y_i$  is the actual observed value,  $\hat{y}_i$  is the predicted value,  $\bar{y}_i$  is the mean of the observed values, and  $n$  is the total number of observations. The  $R^2$  value ranges from  $-\infty$  to 1, where a value closer to 1 indicates a better fit of the model.

The mean absolute percentage error quantifies the relative error in predictions as a percentage. It is given by:

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (S10)$$

where the terms have the same meanings as in the  $R^2$  equation. MAPE expresses the error as a percentage, making it useful for comparing models across different datasets.

The root mean squared error represents the average squared difference between actual and predicted values. It is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (S11)$$

where the terms have the same meanings as in the  $R^2$  equation. A lower MSE value indicates better predictive accuracy.

To compare models effectively using the three evaluation criteria, a single composite metric can be constructed by normalizing and combining these values as follows:

$$Composite\ score = R^2 - \frac{MAPE}{\max(MAPE)} - \frac{RMSE}{\max(RMSE)} \quad (S12)$$

MAPE and RMSE are normalized by their maximum values across models. Higher composite score indicates better model performance.

### 5.3 Feature selection

The first method used to assess feature importance is Pearson correlation. This approach evaluates the linear relationship between each feature ( $X_i$ ) and the target variable ( $y$ ). The second method leverages a Random Forest Regressor to determine feature importance. The model is trained with 1000 trees, and the importance values are extracted from the trained model. Random Forest evaluates feature importance based on how frequently a feature is used to split data across trees and how much it reduces impurity in predictions. To combine the insights from both methods, the final importance score is obtained by averaging the normalized importance values from the Pearson correlation and Random Forest approaches. A predefined importance threshold of 0.005 is applied to filter out less significant features. Any feature with an overall importance score greater than 0.005 is retained for next step model training, testing, and prediction.



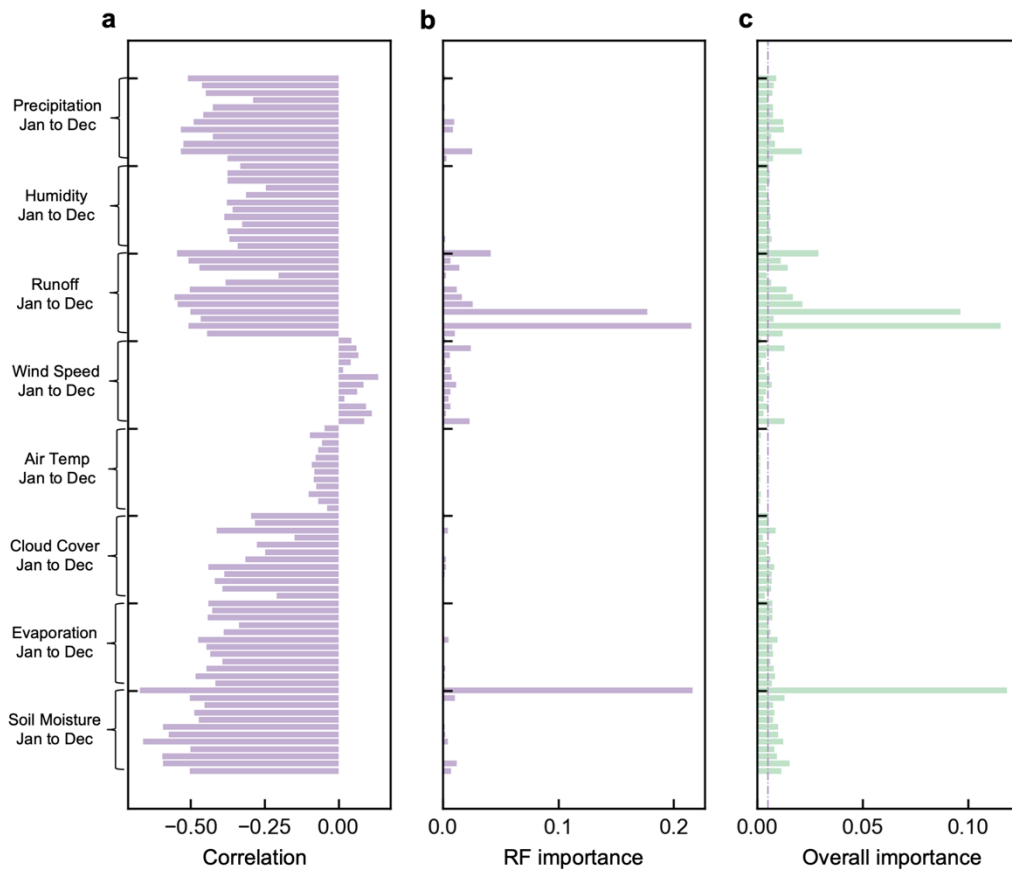


Figure S10. Feature importance (a) correlation, (b) random forest importance metrics, and (c) combined overall importance.

#### 5.4 Model training, testing, and prediction

Irrigation amounts from 2000 to 2014 were used for training, while amounts from 2015 to 2023 served as the testing dataset. Predictions were made for the years 2024 to 2049. Grid search was implemented to optimize the model parameters<sup>20</sup>. The supplementary figures below illustrate the models' performance on the testing dataset, and the evaluation criteria are recorded in Table S4.

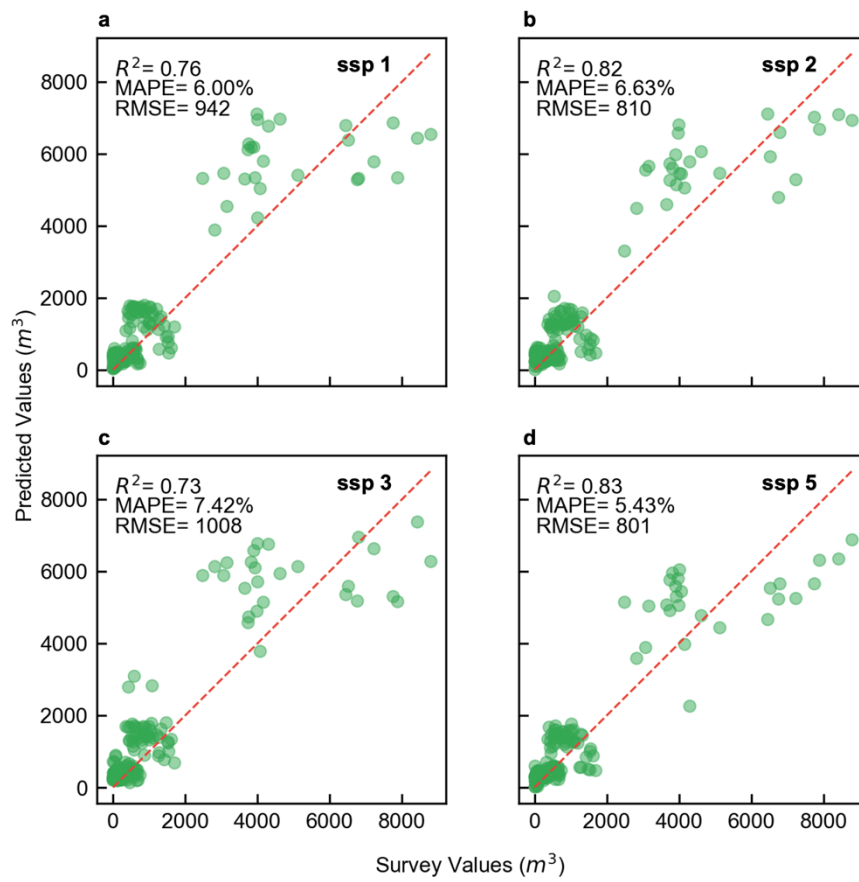


Figure S11. KNN Model performance on testing dataset under (a) SSP1-2.6, (b) SSP2-4.5, (c) SSP3-7.0, and (d) SSP5-8.5.

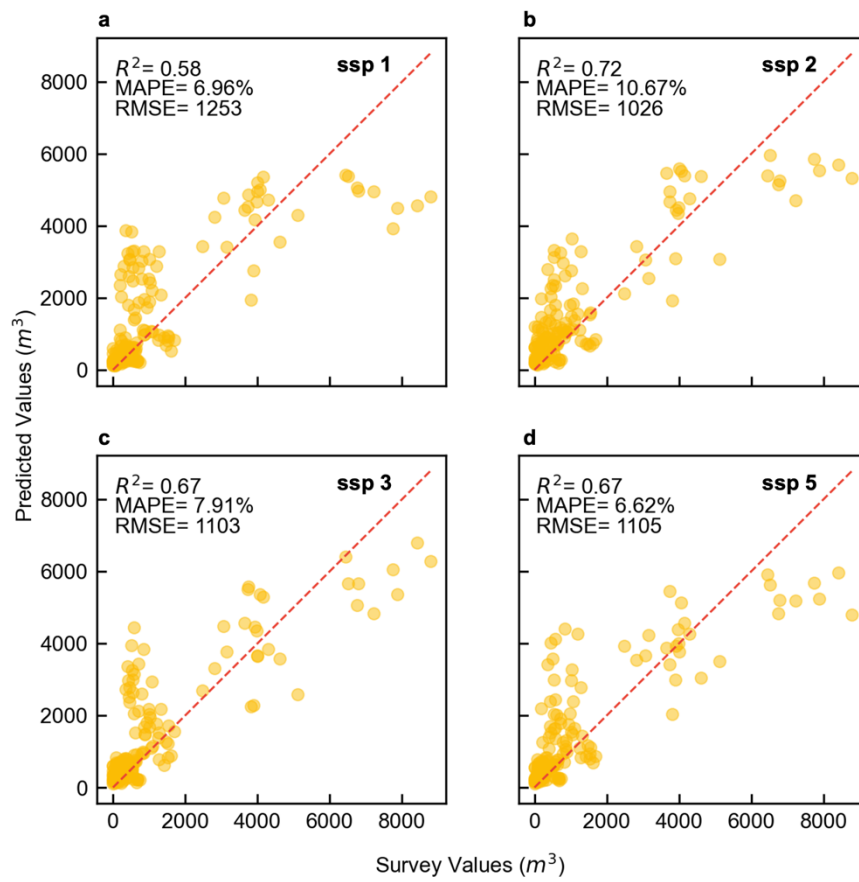


Figure S12. RF Model performance on testing dataset under (a) SSP1-2.6, (b) SSP2-4.5, (c) SSP3-7.0, and (d) SSP5-8.5.

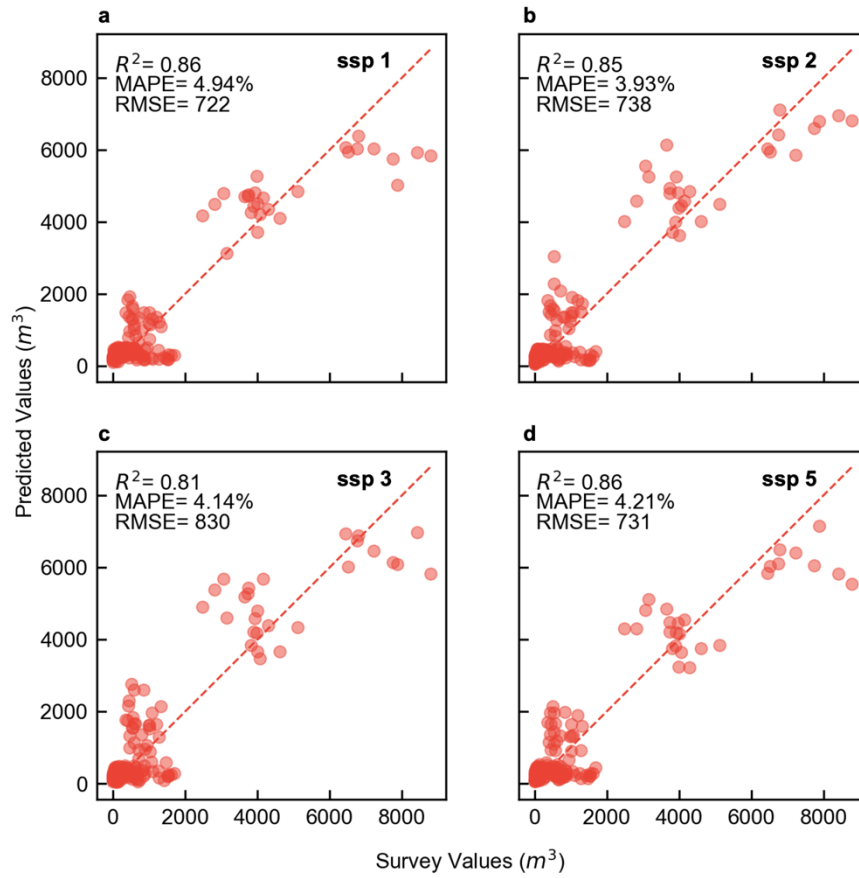


Figure S13. MLP Model performance on testing dataset under (a) SSP1-2.6, (b) SSP2-4.5, (c) SSP3-7.0, and (d) SSP5-8.5.

Supplementary Table S4. Summary of model performance on testing dataset.

| Criteria | Scenario | Ensemble Model | KNN  | RF   | MLP        |
|----------|----------|----------------|------|------|------------|
| $R^2$    | SSP1     | <b>0.90</b>    | 0.76 | 0.58 | 0.86       |
|          | SSP2     | <b>0.88</b>    | 0.82 | 0.72 | 0.85       |
|          | SSP3     | <b>0.86</b>    | 0.73 | 0.67 | 0.81       |
|          | SSP5     | <b>0.90</b>    | 0.83 | 0.67 | 0.86       |
| MAPE     | SSP1     | <b>4.3</b>     | 6.0  | 7.0  | 4.9        |
|          | SSP2     | <b>4.5</b>     | 6.6  | 10.7 | 3.9        |
|          | SSP3     | 5.4            | 7.4  | 7.9  | <b>4.1</b> |
|          | SSP5     | 4.9            | 5.4  | 6.6  | <b>4.2</b> |
| RMSE     | SSP1     | <b>598</b>     | 942  | 1253 | 722        |
|          | SSP2     | <b>674</b>     | 810  | 1026 | 738        |
|          | SSP3     | <b>724</b>     | 1008 | 1103 | 830        |
|          | SSP5     | <b>621</b>     | 891  | 1105 | 731        |

|                    |      |              |       |       |       |
|--------------------|------|--------------|-------|-------|-------|
| Composite<br>Score | SSP1 | <b>0.02</b>  | -0.55 | -1.07 | -0.17 |
|                    | SSP2 | <b>-0.08</b> | -0.44 | -1.10 | -0.10 |
|                    | SSP3 | <b>-0.22</b> | -0.77 | -0.95 | -0.24 |
|                    | SSP5 | <b>-0.05</b> | -0.39 | -0.83 | -0.12 |

\* The best performance was marked in bold

## 6 Weather data interpretation

Figure 14 presents the national average soil moisture for June, July, and August. These three months were selected because they represent the peak irrigation season and highest water demand period for cotton production. We observed that irrigation was notably high in the years 2011, showing a clear increase from historical averages. Upon examining the soil moisture patterns, we found that the average soil moisture in the year was relatively low compared to their respective decades. This suggests that the declining soil moisture trend during peak summer months may be attributed to changing climatic conditions, potentially requiring higher irrigation inputs to maintain adequate soil water content for crop production.

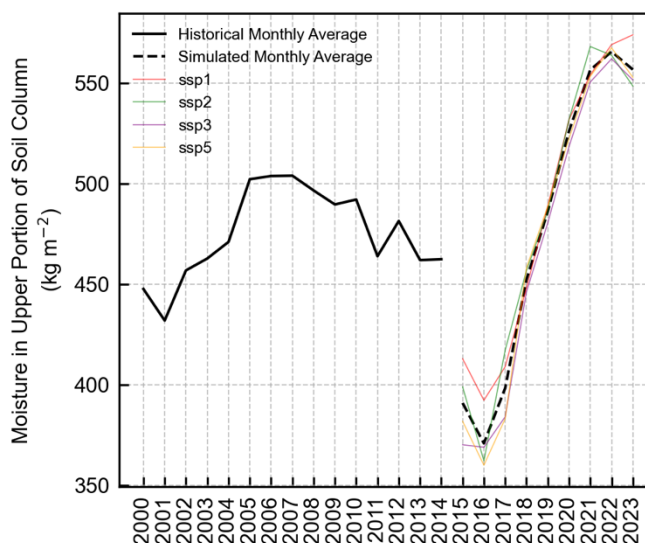


Figure S14. National average soil moisture in June, July, and August.

## 7 Uncertainty in each states' production percentage

As described in Equation (5) in main text, the national unit water consumption is calculated as a weighted average based on each state's production. Figure S15 illustrates the variation in state-level cotton production percentages from 2000 to 2023, introducing uncertainty in the calculation of national average water consumption. To address this, the minimum and maximum cotton production percentages for each state were extracted to define the data range, which served as the basis for generating synthetic samples. A rejection sampling method was employed to generate valid synthetic samples. This process involved iteratively drawing random values from uniform

distributions defined by each state's minimum and maximum production percentage values. Samples were accepted only if the sum of production percentages for 17 states was approximately equal to one (with a tolerance of 0.0001), ensuring that the total cotton production percentage across the 17 states equaled 100%. A total of 1,000 group of samples were generated using this approach. Each group covers production percentages for 17 states.

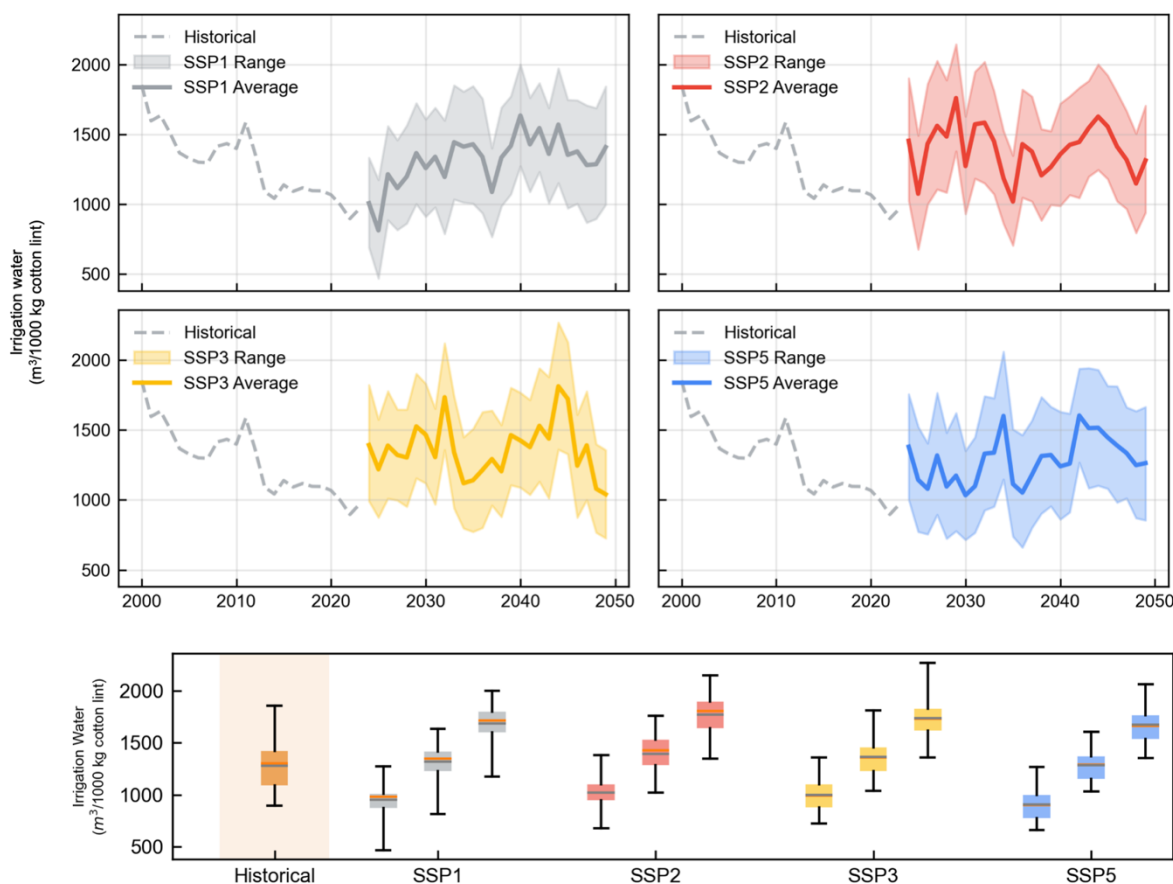


Figure S15. National average irrigation water predictions with uncertainty.

## 8 State-level LCA Results

A systematic Life Cycle Assessment calculation using the most recent historical data from 2023 has been implemented for cotton production across 17 U.S. states. The analysis employs a component-wise multiplication approach where each agricultural input is multiplied by its corresponding environmental impact factor. The methodology employs impact factors sourced primarily from the GREET model, which are shown in "energy analysis summary" excel sheet in the open repository. State-specific electricity grid factors are incorporated, ranging from 0.25 kg CO<sub>2</sub>-eq/kWh in California to 0.90 kg CO<sub>2</sub>-eq/kWh in Missouri, reflecting regional differences in electricity generation sources. The calculation covers three environmental impact categories including greenhouse gas emissions (kg CO<sub>2</sub>-eq), energy consumption (MJ), and water usage (m<sup>3</sup>),

528 per 1000 kg of cotton lint produced. The electrification scenario analysis employs a precise 4.4  
529 kWh/L conversion ratio to replace diesel consumption with electricity, with results calculated  
530 using state-specific electricity factors to account for regional grid variations. All calculation details  
531 and results could be found from the “Cotton\_LCA\_17\_States” excel sheet in the open repository.

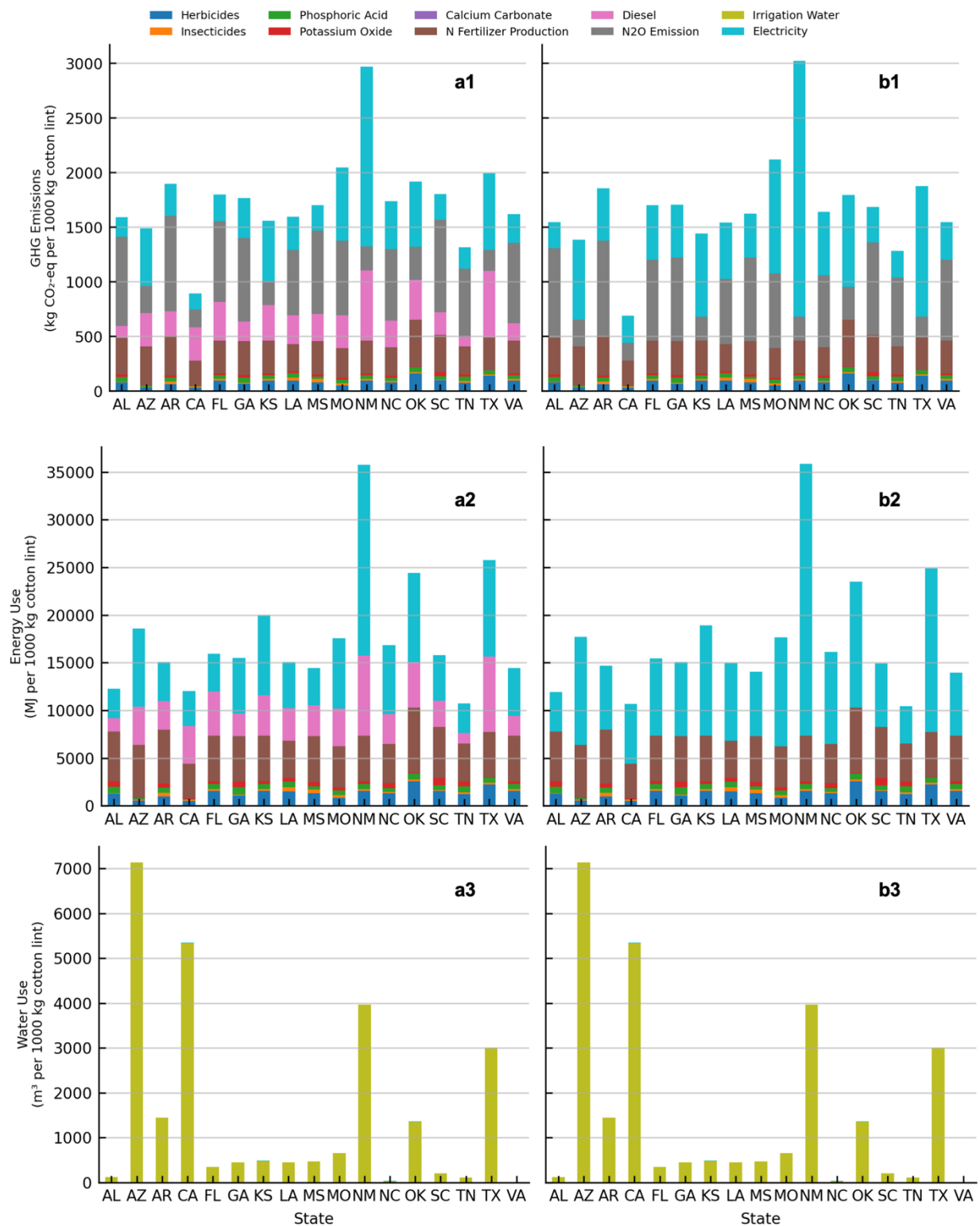


Figure S16. State-level life cycle assessment results based on data from 2023 for cotton production. (a) Baseline scenario reflecting current production practices; (b) Electrification scenario substituting diesel fuel with electricity at a conversion rate of 4.4 kWh per liter of diesel.



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